

Predicting Student Persistence and Dropout

An interpretable machine-learning analysis using socio-academic data after two semesters of study

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0.8142	0.7869 +/- 0.0129	4,424
Held-out test F1	Nested-CV F1	Student records

1. Overview

The analysis estimates whether a student belongs to the dropout class. The source dataset contains three outcomes: Graduate, Enrolled, and Dropout. For this notebook, Dropout is the positive class, while Graduate and Enrolled are grouped as non-dropout.

The predictors cover enrollment history, demographic and socioeconomic characteristics, macroeconomic indicators, and academic performance. Since the final feature set uses second-semester results, the model is intended for risk monitoring after two semesters. It is not an enrollment-time early-warning model.

2. Data and preparation

The UCI dataset contains 4,424 records and 36 predictors. The published observations contain no missing values, so no artificial missingness is introduced. Imputation remains part of the pipeline so that future incomplete records can still be processed.

Completion ratios and engagement indicators are derived from curricular-unit counts. Redundant first-semester measures and variables with very limited variation are removed, while second-semester performance is retained. Numerical variables are scaled, skewed age values are log-transformed, and nominal variables are one-hot encoded.

3. Model selection

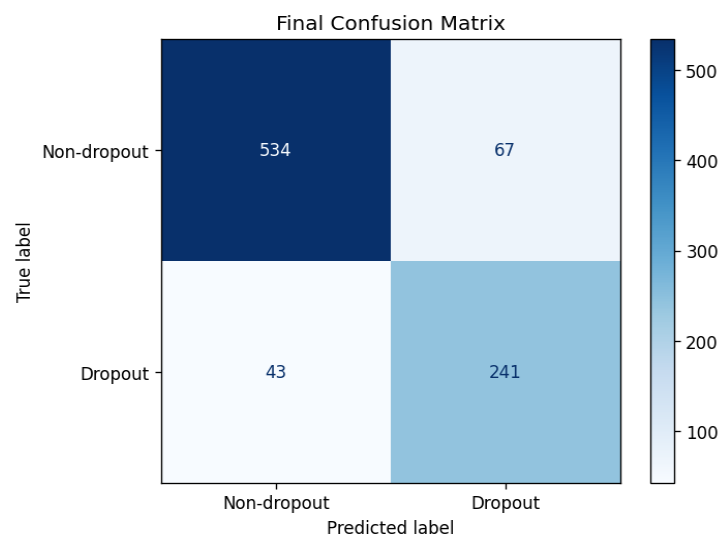
The search compares logistic regression, perceptron, k-nearest neighbors, and random forest classifiers. It also considers no resampling, random over-sampling, random under-sampling, and SMOTE, together with optional PCA or LDA dimensionality reduction.

Nested cross-validation separates model selection from evaluation within the training split. A three-fold inner search samples 200 combinations, and five outer folds estimate the performance of the complete search procedure. The mean outer-fold F1-score is 0.7869 with a standard deviation of 0.0129.

L1-regularized logistic regression was selected in three of the five outer folds, while random forest was selected in two. The final candidate combines random over-sampling with L1-regularized logistic regression because it offers competitive performance, explicit treatment of class imbalance, and coefficients that can be inspected. A final five-fold search selected $C = 1$.

4. Test results

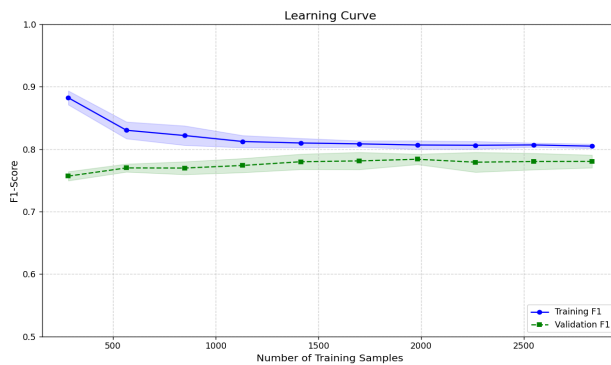
Measure	Score	Interpretation
Training F1	0.8003	Fit on the complete training split
Final CV F1	0.7865	Five-fold tuning of the selected pipeline
Held-out test F1	0.8142	Reported once after model selection
Generalization gap	-0.0138	No evident overfitting in this run



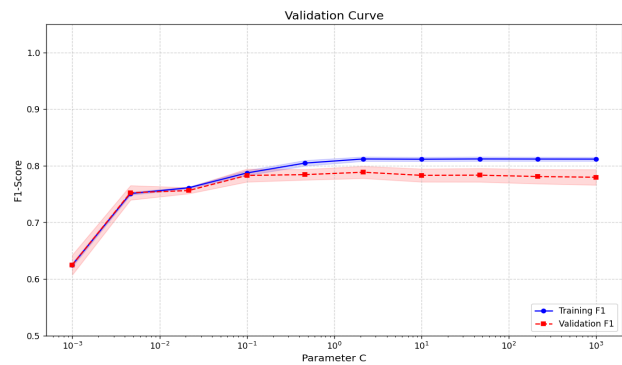
Final confusion matrix for non-dropout and dropout predictions on the held-out test set.

5. Generalization checks

The learning curve shows the training and validation scores moving closer as the training set grows. The validation curve shows that performance is reasonably stable around the selected regularization range. Together with the held-out result, these plots provide no clear evidence of overfitting, although performance should still be confirmed on data from another institution or cohort.



Learning curve



Validation curve for C

6. Interpretation and limitations

The largest positive logistic-regression coefficients occur in categories related to previous qualification, parental education and occupation, and degree program. These values are model associations, not causal effects. One-hot encoded categories must also be interpreted relative to their omitted reference category.

The binary target treats currently enrolled and graduated students as one non-dropout class. This makes the task useful for identifying dropout risk, but it does not distinguish persistence from completed graduation. The data comes from one higher-education institution, and the use of second-semester performance limits how early the predictions can be made.

7. Conclusion

The final pipeline provides a compact and interpretable benchmark for dropout-risk classification after two semesters. Its held-out F1-score of 0.8142 is consistent with the cross-validation results, while the coefficient analysis offers a transparent view of the variables used by the model. External validation would be the most useful next step.

References

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>

Repository: https://github.com/Vittoriacrugnolaa/University_projects/tree/main/Student_Dropout